

7.5.4 Rotation of Axes; Second Degree Equations

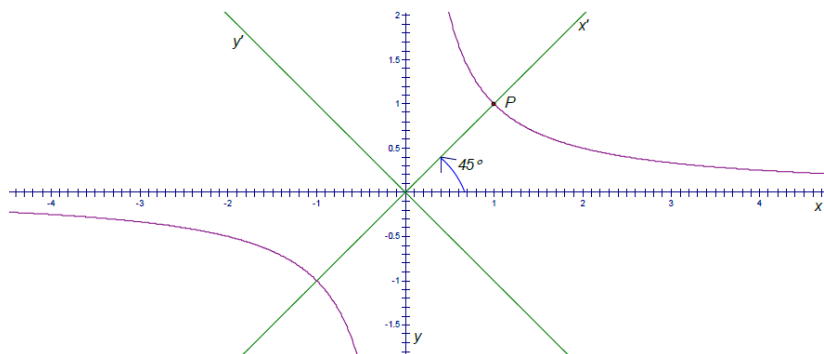
So far, all major/minor axes, transverse/conjugate axes, and directrices have been vertical or horizontal. We now consider situations where this is not the case.

Example 1:

Consider $xy = 1 \Leftrightarrow y = \frac{1}{x}$ (it is a hyperbola). Rotating the x and y axes by 45° makes it easier to analyze this hyperbola.

In the picture the hyperbola is in purple, and the new x and y axes (after the rotation) are in green. This hyperbola appears to have been *tilted*.

Note: P has coordinates with respect to the xy -plane and with respect to the $x'y'$ -plane. Can you figure out P 's coordinates with respect to each plane?



Coordinates of P with respect to the xy -plane: $(1,1)$

Coordinates of P with respect to the $x'y'$ -plane: $(\sqrt{2}, 0)$

Note: The key to figuring out P 's coordinates with respect to the $x'y'$ -plane is recognizing its distance from the origin. We will generalize this idea in the next example.

General Idea:

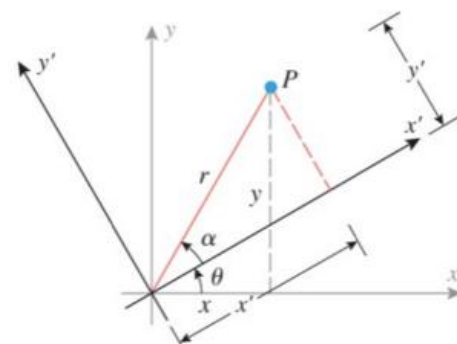
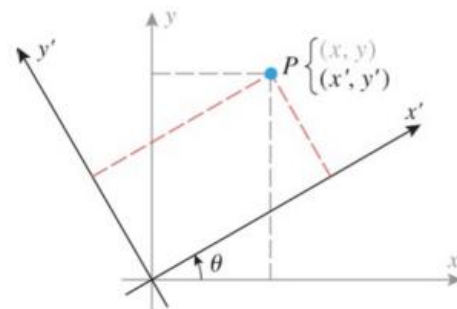
If a point P has xy -coordinates of (x, y) then after rotating the axes by θ we can find the $x'y'$ -coordinates by using the following equations:

$$\begin{aligned} x &= r \cos(\theta + \alpha) & y &= r \sin(\theta + \alpha) \\ x' &= r \cos(\alpha) & y' &= r \sin(\alpha) \end{aligned}$$

Using identities for $\cos(\theta + \alpha)$ & $\sin(\theta + \alpha)$ we can see that:

$$\begin{aligned} x &= r \cos(\theta) \cos(\alpha) - r \sin(\theta) \sin(\alpha) \\ y &= r \sin(\theta) \cos(\alpha) + r \sin(\alpha) \cos(\theta) \end{aligned}$$

Then by substituting in the in the equations for x' & y' we have:



$$\begin{cases} x = x' \cos(\theta) - y' \sin(\theta) \\ y = x' \sin(\theta) + y' \cos(\theta) \end{cases}$$

We can also develop equations for x' & y' as follows:

Multiply the first equation by $\cos(\theta)$ and the second by $\sin(\theta)$:

$$\begin{cases} x \cos(\theta) = x' \cos^2(\theta) - y' \sin(\theta) \cos(\theta) \\ y \sin(\theta) = x' \sin^2(\theta) + y' \sin(\theta) \cos(\theta) \end{cases} \rightarrow \begin{cases} x \cos(\theta) = x' \cos^2(\theta) - y' \sin(\theta) \cos(\theta) \\ y \sin(\theta) = x' \sin^2(\theta) + y' \sin(\theta) \cos(\theta) \end{cases}$$

$$\boxed{x \cos(\theta) + y \sin(\theta) = x'}$$

Similarly if we multiply the first equation by $-\sin(\theta)$ and the second by $\cos(\theta)$:

$$\begin{cases} -x \sin(\theta) = -x' \sin(\theta) \cos(\theta) + y' \sin^2(\theta) \\ y \cos(\theta) = x' \sin(\theta) \cos(\theta) + y' \cos^2(\theta) \end{cases} \rightarrow \begin{cases} -x \sin(\theta) = -x' \sin(\theta) \cos(\theta) + y' \sin^2(\theta) \\ y \cos(\theta) = x' \sin(\theta) \cos(\theta) + y' \cos^2(\theta) \end{cases}$$

$$\boxed{-x \sin(\theta) + y \cos(\theta) = y'}$$

Theorem: Let θ be the angle of rotation of the x and y axes. If (x, y) is a point in the original system and (x', y') is the same point in the new system, then

$$\begin{aligned} x' &= x \cos(\theta) + y \sin(\theta) & x &= x' \cos(\theta) - y' \sin(\theta) \\ y' &= -x \sin(\theta) + y \cos(\theta) & y &= x' \sin(\theta) + y' \cos(\theta) \end{aligned}$$

Note: We can see that this theorem does indeed work if we consider example 1.

Example 2:

Find the new coordinates of the point $(2, 4)$ if the new coordinates axes are rotated through an angle of $\theta = 30^\circ$.

Using the equations above we have:

$$x' = 2 \cos(30^\circ) + 4 \sin(30^\circ) \rightarrow x' = 2 \left(\frac{\sqrt{3}}{2} \right) + 4 \left(\frac{1}{2} \right) = \sqrt{3} + 2$$

$$y' = -2 \sin(30^\circ) + 4 \cos(30^\circ) \rightarrow y' = -2 \left(\frac{1}{2} \right) + 4 \left(\frac{\sqrt{3}}{2} \right) = -1 + 2\sqrt{3}$$

Therefore the coordinates of the new point are $(\sqrt{3} + 2, -1 + 2\sqrt{3})$

Example 3:

Express $xy = 1$ in terms of new $x'y'$ -coordinates by rotating the axes through an angle of $\theta = 45^\circ$

Using the equations above we can replace x and y and obtain:

$$\left[x' \cos(45^\circ) - y' \sin(45^\circ) \right] \left[x' \sin(45^\circ) + y' \cos(45^\circ) \right] = 1$$

$$\Leftrightarrow x'^2 \cos(45^\circ) \sin(45^\circ) - y'^2 \sin(45^\circ) \cos(45^\circ)$$

$$\Leftrightarrow x'^2 \left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) - y'^2 \left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = 1$$

$$\Leftrightarrow \frac{x'^2}{2} - \frac{y'^2}{2} = 1$$

Theorem: If a conic is given by the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

is such that $B \neq 0$, and if an $x'y'$ -coordinate system is obtained by rotating the xy -axes through an angle θ satisfying

$$\cot(2\theta) = \frac{A-C}{B}$$

then, in $x'y'$ -coordinates, the given conic will have the form

$$A'x'^2 + C'y'^2 + D'x' + E'y' + F' = 0$$

Proof: see textbook pages 750&751.

Note: We will always choose θ to be acute.

Example 4: Identify and sketch the conic given by the equation

$$x^2 + \sqrt{3}xy + 2y^2 - 10 = 0$$

Since the equation contains an xy -term we know that it is a rotated conic.

We begin by attempting to find the angle of rotation using the previous theorem.

$$\cot(2\theta) = \frac{1-2}{\sqrt{3}} = -\frac{\sqrt{3}}{3} \rightarrow 2\theta = \frac{2\pi}{3} \Leftrightarrow \theta = \frac{\pi}{3}$$

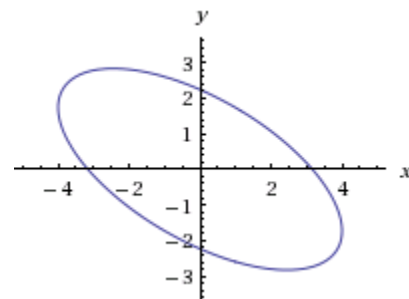
In order to convert the equation we need to find the value of x and y using the first theorem from this lesson.

$$\begin{aligned} \begin{aligned} x &= x' \cos(\theta) - y' \sin(\theta) \\ y &= x' \sin(\theta) + y' \cos(\theta) \end{aligned} &\Leftrightarrow \begin{aligned} x &= x' \cos\left(\frac{\pi}{3}\right) - y' \sin\left(\frac{\pi}{3}\right) \\ y &= x' \sin\left(\frac{\pi}{3}\right) + y' \cos\left(\frac{\pi}{3}\right) \end{aligned} &\Leftrightarrow \begin{aligned} x &= \frac{1}{2}x' - \frac{\sqrt{3}}{2}y' \\ y &= \frac{\sqrt{3}}{2}x' + \frac{1}{2}y' \end{aligned} &\Leftrightarrow \begin{aligned} x &= \frac{1}{2}(x' - \sqrt{3}y') \\ y &= \frac{1}{2}(\sqrt{3}x' + y') \end{aligned} \end{aligned}$$

We can use these equations to convert the given equation:

$$\begin{aligned}
 & \left(\frac{1}{2}(x' - \sqrt{3}y') \right)^2 + \sqrt{3} \frac{1}{4}(x' - \sqrt{3}y')(\sqrt{3}x' + y') + 2 \left(\frac{1}{2}(\sqrt{3}x' + y') \right)^2 - 10 = 0 \\
 \Leftrightarrow & \frac{1}{4}(x'^2 - 2\sqrt{3}x'y' + 3y'^2) + \frac{\sqrt{3}}{4}(\sqrt{3}x'^2 + x'y' - 3x'y' - \sqrt{3}y'^2) + \frac{1}{2}(3x'^2 + 2\sqrt{3}x'y' + y'^2) = 10 \\
 \Leftrightarrow & (x'^2 - 2\sqrt{3}x'y' + 3y'^2) + \sqrt{3}(\sqrt{3}x'^2 - 2x'y' - \sqrt{3}y'^2) + 2(3x'^2 + 2\sqrt{3}x'y' + y'^2) = 40 \\
 \Leftrightarrow & x'^2 - 2\sqrt{3}x'y' + 3y'^2 + 3x'^2 - 2\sqrt{3}x'y' - 3y'^2 + 6x'^2 + 4\sqrt{3}x'y' + 2y'^2 = 40 \\
 \Leftrightarrow & 10x'^2 + 2y'^2 = 40 \\
 \Leftrightarrow & \frac{x'^2}{4} + \frac{y'^2}{20} = 1
 \end{aligned}$$

This is an ellipse with a center of $(0,0)$. Note that $c^2 = 20 - 4 = 16 \Leftrightarrow c = 4$, so the coordinates of the foci (with respect to the new axes) are $(0, \pm 4)$.



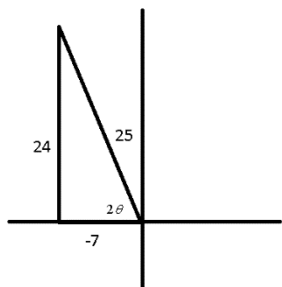
Example 5: Consider the conic given by the equation
 $153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0$

Find equations for x and y needed to express this conic with respect to the $x'y'$ axes.

We begin by attempting to find the angle of rotation using the previous theorem.

$$\cot(2\theta) = \frac{153 - 97}{-192} = -\frac{56}{192} = -\frac{7}{24}$$

Since this equation is not easy to solve for θ we will instead use a triangle and some trig. identities:



$$\begin{aligned}
 \cos^2(\theta) &= \frac{1}{2}[1 + \cos(2\theta)] & \Leftrightarrow & \cos^2(\theta) = \frac{1}{2}\left[1 - \frac{7}{25}\right] & \Leftrightarrow & \cos^2(\theta) = \frac{18}{50} & \Leftrightarrow & \cos(\theta) = \frac{3}{5} \\
 \sin^2(\theta) &= \frac{1}{2}[1 - \cos(2\theta)] & \Leftrightarrow & \sin^2(\theta) = \frac{1}{2}\left[1 + \frac{7}{25}\right] & \Leftrightarrow & \sin^2(\theta) = \frac{32}{50} & \Leftrightarrow & \sin(\theta) = \frac{4}{5}
 \end{aligned}$$

Notice that now we do not need to find the value of θ and can begin to find the value of x and y :

$$\begin{aligned}
 x &= x' \cos(\theta) - y' \sin(\theta) & \Leftrightarrow & x = \frac{3}{5}x' - \frac{4}{5}y' & \Leftrightarrow & x = \frac{1}{5}(3x' - 4y') \\
 y &= x' \sin(\theta) + y' \cos(\theta) & \Leftrightarrow & y = \frac{4}{5}x' + \frac{3}{5}y' & \Leftrightarrow & y = \frac{1}{5}(4x' + 3y')
 \end{aligned}$$

From here we could continue and rewrite the given equation with respect to the $x'y'$ -axes.

Example 6: Show that the given equation is a parabola, then find its vertex, focus, and directrix in terms of the xy -axes and sketch a graph.

$$\frac{1}{2}(x^2 + y^2) - 2(x + y - 1) - xy = 0$$

Since the equation contains an xy -term we know that it is a rotated conic.

We begin by attempting to find the angle of rotation using the previous theorem.

$$\cot(2\theta) = \frac{\frac{1}{2} - \frac{1}{2}}{-1} = 0 \Leftrightarrow 2\theta = \frac{\pi}{2} \Leftrightarrow \theta = \frac{\pi}{4}$$

Since this equation is not easy to solve for θ we will instead use a triangle and some trig. identities:

$$\begin{aligned} \begin{aligned} x &= x' \cos(\theta) - y' \sin(\theta) \\ y &= x' \sin(\theta) + y' \cos(\theta) \end{aligned} &\Leftrightarrow \begin{aligned} x &= x' \cos\left(\frac{\pi}{4}\right) - y' \sin\left(\frac{\pi}{4}\right) \\ y &= x' \sin\left(\frac{\pi}{4}\right) + y' \cos\left(\frac{\pi}{4}\right) \end{aligned} &\Leftrightarrow \begin{aligned} x &= \frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y' \\ y &= \frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y' \end{aligned} &\Leftrightarrow \begin{aligned} x &= \frac{\sqrt{2}}{2}(x' - y') \\ y &= \frac{\sqrt{2}}{2}(x' + y') \end{aligned} \end{aligned}$$

We can use these equations to convert the given equation

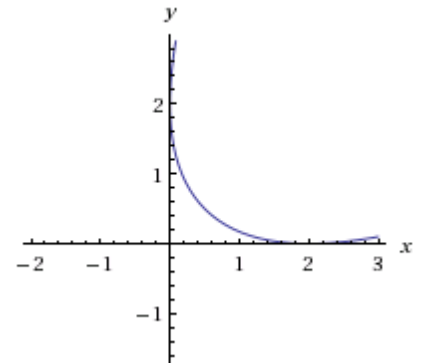
$$\frac{1}{2} \left(\left(\frac{\sqrt{2}}{2}(x' - y') \right)^2 + \left(\frac{\sqrt{2}}{2}(x' + y') \right)^2 \right) - 2 \left(\frac{\sqrt{2}}{2}(x' - y') + \frac{\sqrt{2}}{2}(x' + y') - 1 \right) - \frac{1}{2}(x' - y')(x' + y') = 0$$

$$\Leftrightarrow \frac{1}{2} \left(\frac{1}{2}(x'^2 - 2x'y' + y'^2 + x'^2 + 2x'y' + y'^2) \right) - 2(\sqrt{2}x' - 1) - \frac{1}{2}x'^2 + \frac{1}{2}y'^2 = 0$$

$$\Leftrightarrow y'^2 - (2\sqrt{2}x' - 2) = 0 \Leftrightarrow y'^2 + 2 = 2\sqrt{2}x' \Leftrightarrow \frac{\sqrt{2}}{4}y'^2 + \frac{\sqrt{2}}{2} = x'$$

This is a parabola with $a = \frac{1}{4p} = \frac{\sqrt{2}}{4} \Leftrightarrow p = \frac{\sqrt{2}}{2}$, $k = 0$, & $h = \frac{\sqrt{2}}{2}$.

Therefore it has vertex $\left(\frac{\sqrt{2}}{2}, 0\right)$, focus $(\sqrt{2}, 0)$, and directrix $x' = 0$ with respect to the $x'y'$ -axes.



To convert these to points/lines in the xy axes we will use the equations from the first theorem:

$$\begin{aligned} \text{Vertex: } \begin{aligned} x &= \frac{\sqrt{2}}{2} \left(\frac{\sqrt{2}}{2} - 0 \right) = \frac{1}{2} \\ y &= \frac{\sqrt{2}}{2} \left(\frac{\sqrt{2}}{2} + 0 \right) = \frac{1}{2} \end{aligned} &\rightarrow \left(\frac{1}{2}, \frac{1}{2} \right) & \text{Focus: } \begin{aligned} x &= \frac{\sqrt{2}}{2} (\sqrt{2} - 0) = 1 \\ y &= \frac{\sqrt{2}}{2} (\sqrt{2} + 0) = 1 \end{aligned} &\rightarrow (1, 1) \end{aligned}$$

Directrix: We have $x' = 0$, so

$$0 = x \cos(\theta) + y \sin(\theta) \Leftrightarrow 0 = x \cos\left(\frac{\pi}{4}\right) + y \sin\left(\frac{\pi}{4}\right) \Leftrightarrow 0 = \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y \Leftrightarrow y = -x$$